

Exam Solution

A

1

$$H = \lambda \begin{bmatrix} R & t \\ \mathbf{0}^T & 1 \end{bmatrix}$$

- 3D: 6 DOF/paramters
- 2D : 3 DOF/paramters

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a)

$$H = \lambda \begin{bmatrix} 1 & 0 & 0 & -10 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} x$$

b)

$$H = \lambda \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x$$

c)

$$H = \lambda \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x$$

3.

Representation	Pros	Cons
Rotation Matrix	Unique, No Singularities	Not Readable
Euler angles	Readable, Minimal Representation	Not Unique, Singularities, Gimbal Lock
Quaternions	Unique, No Singularities	Not (easily) readable

Unique: Ways of representing the Rotation (For Euler Angles: Multiple sets of angles can represent same orientation)

- **Rotation Matrices**: Exchange Format
- **Euler Angles**: Visualisation
- **Quaternions**: State Estimation
- **Angle-Axis**: Readability and Link to Quaternions

4.

- Can simplify mathematical expressions
- Can Model points at infinity with finite coordinates

- $x_{\infty} : \mathbf{x}_{\infty} = \begin{bmatrix} u \\ v \\ 0 \end{bmatrix}$

- Infinitively distant line (**ideal line**)

- Can be seen as the **horizon**

- $l_{\infty} : \mathbf{l}_{\infty} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

- Easy chaining and inversion of transformations

-

- Chaining transformations via matrix products (not commutative)

$$X' = H_1 H_2 X$$

- $\neq H_2 H_1 X$

- Inverting a transformation

$$X' = HX$$

- $X = H^{-1}X'$

- Equivalence up to scale

- $\mathbf{x} = \lambda \mathbf{x}$

- Uses extra dimension

- $\underset{\text{Euclidian}}{\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}} \Rightarrow \underset{\text{homogeneous}}{\mathbf{x} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}}$

5.

a.

$$l = x \wedge y : \quad \mathbf{l} = \mathbf{x} \times \mathbf{y}$$

$$\mathbf{l}_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix} \times \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ -13 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} -2/13 \\ -5/13 \end{bmatrix}$$

b.

$$\mathbf{x} = \mathbf{l} \times \mathbf{m}$$

$$x_3 = \mathbf{l}_0 \times \mathbf{l}_1$$

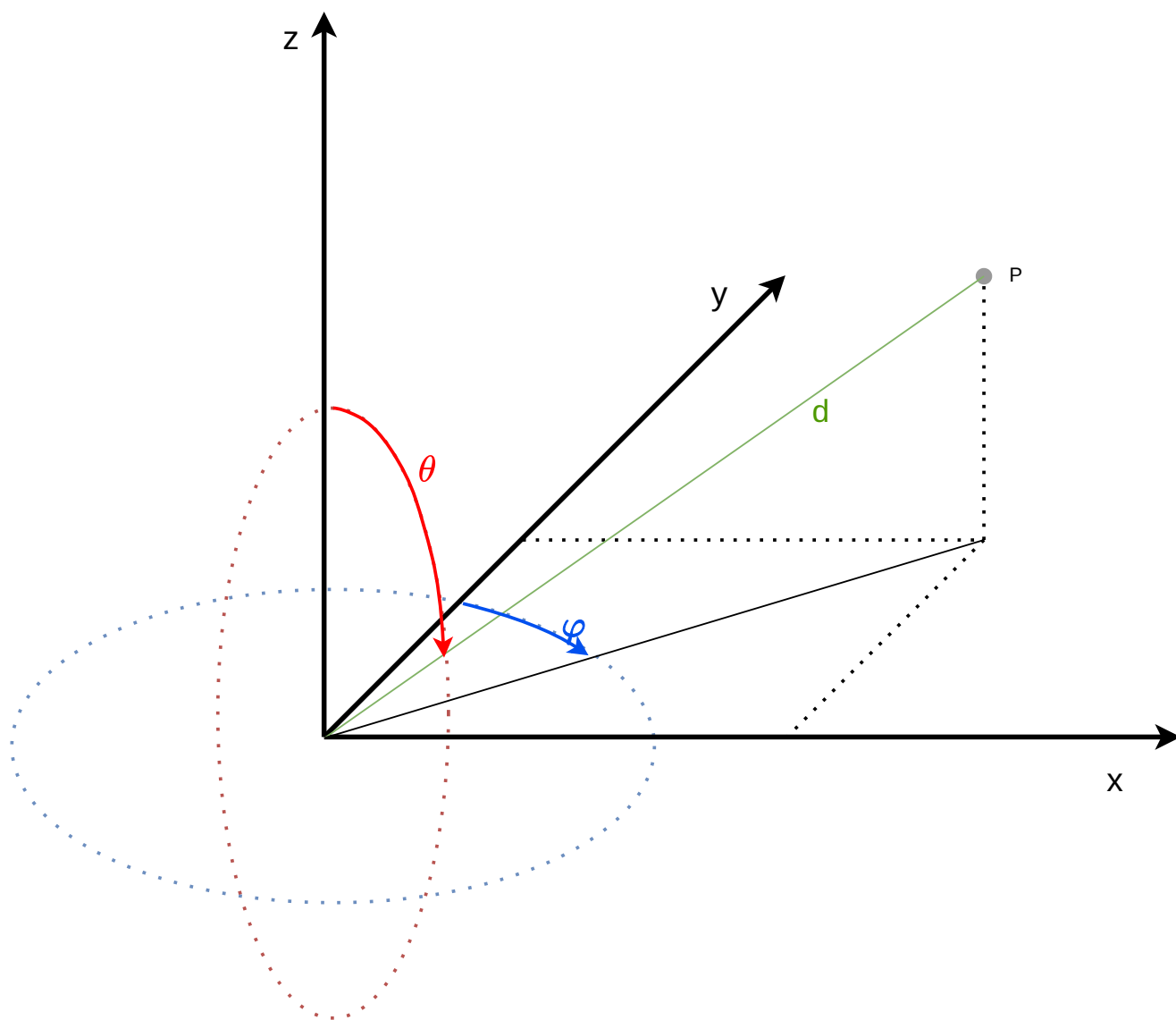
$$x_3 = \begin{bmatrix} -1 \\ -1 \\ 6 \end{bmatrix} \times \begin{bmatrix} 2 \\ 5 \\ -13 \end{bmatrix}$$

$$= \begin{bmatrix} -17 \\ -1 \\ -3 \end{bmatrix}$$

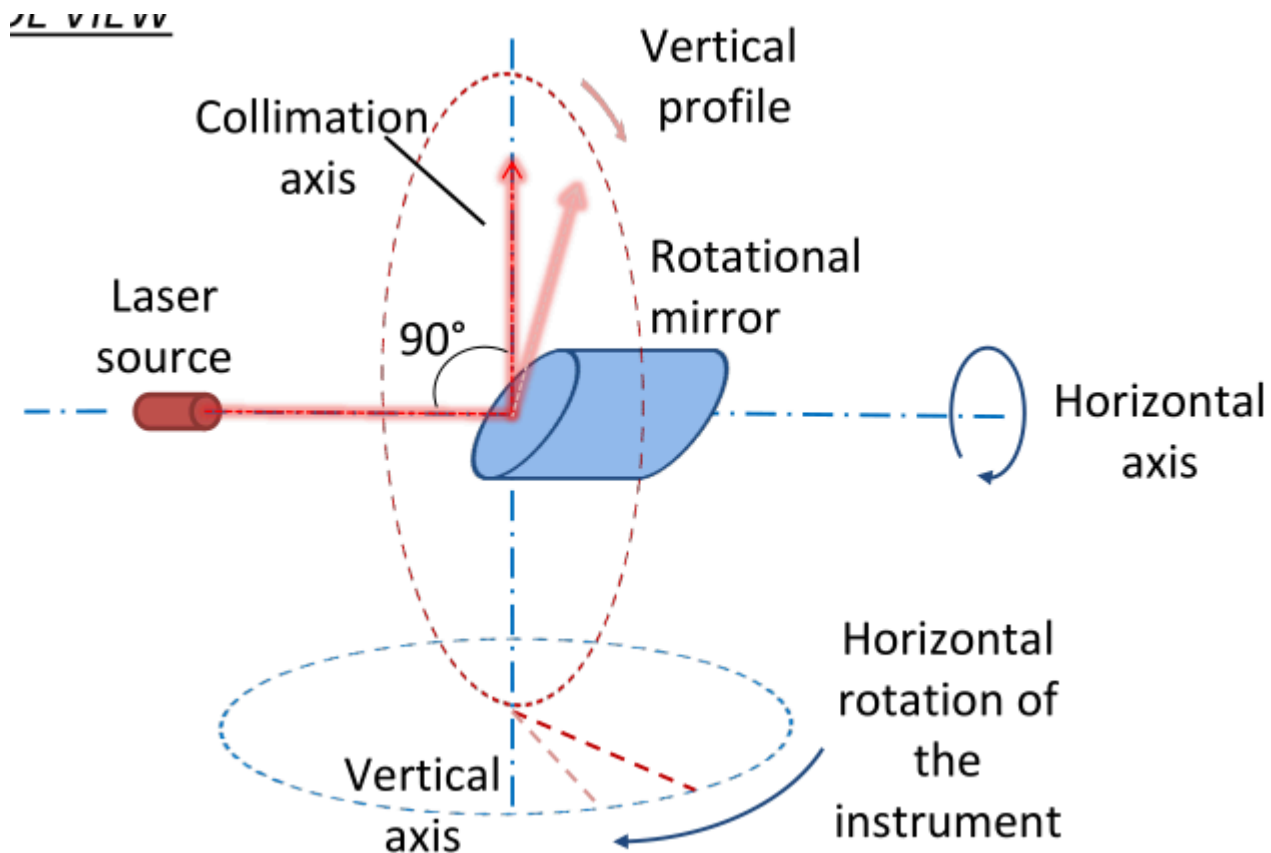
$$\mathbf{x} = \begin{bmatrix} 17/3 \\ 1/3 \end{bmatrix}$$

B

6.



7.



Rotation around Horizontal axis is turning around Collimation axis

-> Vertical Profile

Horizontal rotation around the vertical axis of this apparatus

8.

3D similarity transformation between laser scanner is called **TLS Registration**.

Afterwards Geo-referencing (transformation to global, earth centered CS) and Projection to given official coordinates (split into position and height)

9.

We need well distributed points

- We can exclude the scale $\lambda = 1$
 - When using the same cameras or distances are given in meters (Same fixed Scale μ)
 - Propagation delay of current atmosphere needs to be known
- Rotation $\varepsilon_y = \varepsilon_x = 0$
 - Often static data acquisition and
 - inclinometer assures levelled reference system

7-1-2=4 DoF

10

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \mathbf{R}_3(\varepsilon_z) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

C

11.

For N:

From Equator $\phi = 0^\circ$ to Pole $\phi = 90^\circ$:

$$N(0^\circ) = a \text{ to } N(90^\circ) = \frac{a}{\sqrt{1-e^2}}$$

For M:

From Equator $\phi = 0^\circ$ to Pole $\phi = 90^\circ$:

$$M(0^\circ) = a(1 - e^2) \text{ to } M(90^\circ) = \frac{a(1-e^2)}{\sqrt{1-e^2}} = \frac{a}{\sqrt{1-e^2}}$$

At Pole:

- $N = M$

Everywhere else:

- $N > M$

At the Equator, M is by a factor $(1 - e^2)$ smaller than N.

12.

$$r = a - E = a - e \cdot a = a(1 - e) = 1100(1 - 0.01) = 1089km$$

There is something wrong, because the distance is far too short. at the same time, this should be the solution

13.

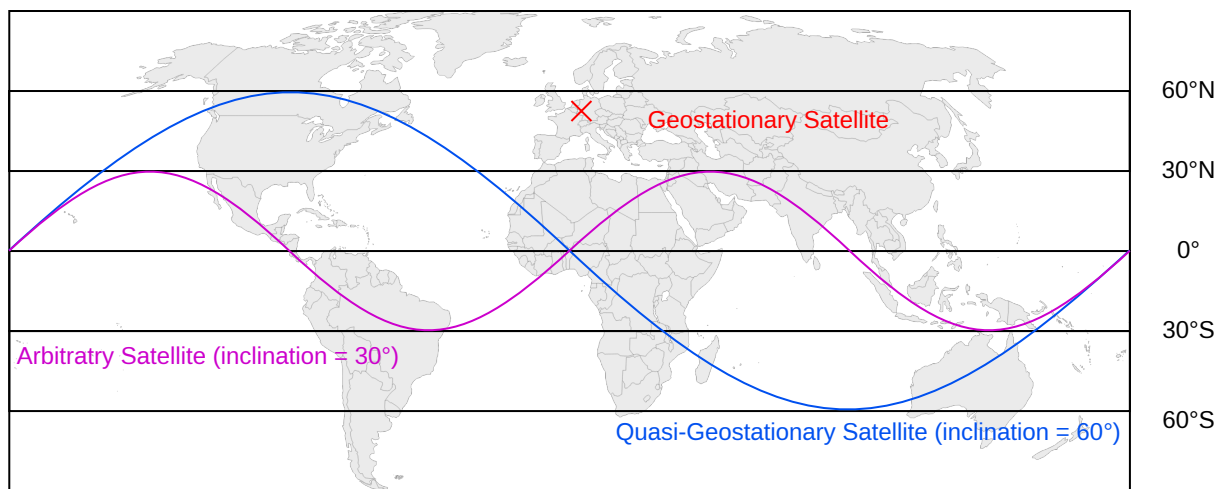
Searching for the Geoid undulation

It is most of the time less than 100m. Hence of the range ~10m

14.

There are 2 independent parameters to describe an ellipsoid. E.g. a and e

15.



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3 Parameters:

$RAAN/\Omega$ - Right ascension of the ascending node

ω Argument of perigee

i - inclination

$$R = R_3(\omega)R_1(i)R_3(\Omega)$$

17.

1. Elliptical Orbits with Earth in one of the foci
2. Line between the central orbit and the satellite sweeps equal areas in equal times
3. Square of the orbital period is proportional to the cube of the orbit's semi-major axis

18.

From orbital theory: rotation rate of $\frac{d\Omega}{dt} = 3.2^\circ d^{-1}$

Due to Earth's flattening (and inhomogeneous density distribution), while Kepler Orbit assumes point masses

Alias Orbital Precession

19.

- Kepler Orbits
- Gravitational Field
- Earth Rotation and Orientation Studies

D

Origin: The center of mass being defined for the whole earth, including oceans and atmosphere.

Scale: The unit of length is the metre (SI).

Orientation: Initially given by the BIH orientation at 1984.0

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- Variations of rotation rate expressed as deviations of UT1 from atomic time scale

- $UT1 - UTC = dUT1$

- Perturbations of nominal rotation rate $1/86400s$

- $\Delta LOD = -\frac{d}{dt}dUT1$

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- $[CRS] = Q(t) \cdot R(t) \cdot W(t) \cdot [TRS]$

- where:

- **Earth's Spin $R(t)$**

- $R(t) = R_z(-ERA)$

- $ERA(T_u) = 2\pi(0.7790572732640 + 1.00273781191135448T_u)$

- MJD of observation epoch 1.9.2020, 16:00UTC $\rightarrow T_u = mjd(UT1epoch) - 51544.5$

- e.g. $mjd(UT1 epoch) = 59093 + 16/24 + dUT1/(24 \cdot 3600)$

- Angle between CEO and TIO: ERA - Earth Rotation Angle

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- **Nutation $Q(t)$**

- $Q(t) = R_z(-E)R_y(-d)R_z(E + s)$

- E, s - other parameters

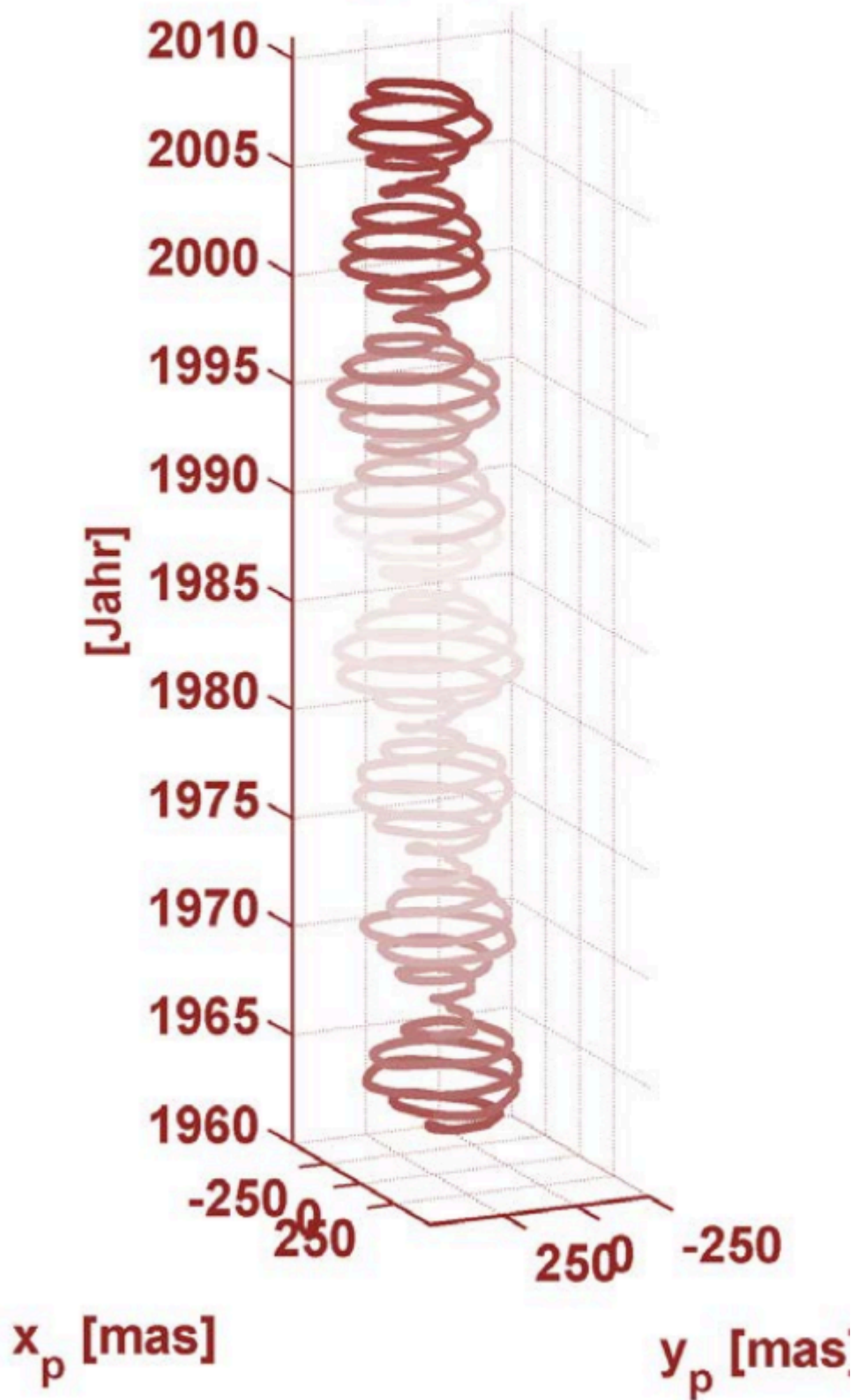
- **Polar motion $W(t)$**

- $W(t) = R_z(-s')R_y(x_p)R_x(y_p)$

- $x_p y_p$ - Pole position in ITRS of CIP

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Polbewegung 1962-2010,7



- Beat
 - Period of 6.3a and peak amplitudes of 9m
- Consists of:
 - Chandler Wobble

- Free oscillation of Earth
- Period ~433 days
- Annual Wobbles
 - Forced oscillation, main driver are annual pressure over continental areas

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- $300mas \rightarrow 300 \cdot 3cm = 900cm = 9m$
- Difference in Celestial frame smaller
 - Mainly reflect nutation and Precession, which is smooth and predictable
 - Polar motion makes apparent motion larger and more irregular (100mas)
- Driven by
 - Tidal Friction: Earth slowing down due to tidal bulge attracte by moon and moving ahead due to earths rotation
 - Core-Mantle Coupling: Rotating liquid outer core couples iwth mantles and slows it down
 - Planetary perturbation from other planets
 - Dynamic Ellipticity -> equatorial bulge enables external forces to generate torque
 - Solar gravitational attratction

E

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- Plate-tectonic stability in Europe
- Practicality for mapping and surveying
 - Projected Coordinates more convenient for planar measurements
- Compatibility with national mapping infrastructure
- Legal and administrative standardisation

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1. Transform WGS84 $\rightarrow$ ETRS89
 - **Transition of Geodetic Datum**
 - Baseline $\Delta X, \Delta Y, \Delta Z$ given in WGS84
 - Apply pure translation (translation parameters should be known)
2. Cartesian $\xrightarrow{GRS80}$ Ellipsoidal
 - **Transition of reference surface**
 - Use Conversion from Cartesian to Ellipsoid (GRS80)

- 2 independent ellipsoidal parameters should be known of the GRS80 (e.g. a, e)

3. Transform latitude and longitude to UTM coordinates

- **Projection**
- Parameters of GRS80 ellipsoid should be known

4. Ellipsoidal height $\xrightarrow{GCG16}$ physical height

- **Transition to reference surface**
- GCG2016 quasi geoid undulations should be known

F

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Disadvantages

- Areas are distorted (esp. close to poles)
- Lengths of meridians are not preserved

Advantages

- Aspect ratios are preserved
- (->) Angles are preserved
- Mercator projection is **conformal**

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$$x_p = r \lambda$$

$$y_p = r \sin \varphi_p$$

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$$\left| \frac{\partial x}{\partial \varphi} \right| \cdot \left| \frac{\partial x}{\partial \lambda} \right| = \left| \frac{\partial X}{\partial \varphi} \right| \cdot \left| \frac{\partial X}{\partial \lambda} \right|$$

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Infinitely small rectangles (Multiplication of two orthogonal sides) on the earth's surface should be equal to the projected rectangles.

Valid for all small Areas-> also for sets of areas

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$$\frac{A_f}{A_{sphere}} = \frac{\left| \frac{\partial x}{\partial \varphi} \right| \cdot \left| \frac{\partial x}{\partial \lambda} \right|}{\left| \frac{\partial X}{\partial \varphi} \right| \cdot \left| \frac{\partial X}{\partial \lambda} \right|}$$

- $(\frac{\partial x}{\partial \varphi}) = \begin{pmatrix} 0 \\ \frac{r}{\cos^2(\varphi)} \end{pmatrix} = (\frac{r}{\cos^2(\varphi)})$
- $(\frac{\partial x}{\partial \lambda}) = r$
- $(\frac{\partial X}{\partial \varphi}) = r$
- $(\frac{\partial X}{\partial \lambda}) = r \cos(\varphi)$

$$\frac{A_f}{A_{sphere}} = \frac{1}{\cos^3(\varphi)}$$

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$$\frac{1}{\cos(50.733992^\circ)} = 3.94412$$